



RESIDENTIAL ENERGY STORAGE / LOW VOLTAGE

VOLTA HOME 16

16.0 kWh LiFePO4 Battery System

A floor-mounted residential battery designed for solar self-consumption, backup power and scalable low-voltage energy storage.

16.0 kWh

RATED ENERGY

51.2 V

RATED VOLTAGE

314 Ah

NOMINAL CAPACITY

6000+

CYCLE LIFE

DESIGNED FOR DAILY ENERGY INDEPENDENCE

- Grade A lithium iron phosphate cells with integrated BMS
- RS232, RS485 and CAN communication interfaces
- Compatible with a broad range of low-voltage inverters
- Up to 15 battery units supported in parallel
- Real-time operating status on the integrated LCD display

REFU VOLT ENERGY STORAGE

M16S314BL-F



APPLICATIONS

Home storage / Solar storage

Backup power / Small commercial

TECHNICAL SPECIFICATIONS

Floor-mounted low-voltage lithium iron phosphate battery system

01	ELECTRICAL
Product model	M16S314BL-F
Cell configuration	16S
Battery chemistry	LiFePO4
Nominal capacity	314 Ah
Rated energy	16.0 kWh
Rated voltage	51.2 V
Charging voltage	58.4 V
Discharge cut-off voltage	42 V
Standard charge current	50 A
Maximum charge current	100 A
Maximum discharge current	200 A

02	OPERATION & COMMUNICATION
Cycle life	>= 6000 cycles
Cycle-life condition	80% DOD, 25 C
Communication	RS232 / RS485 / CAN
Charging temperature	0 to 60 C
Discharge temperature	-10 to 65 C
User display	Integrated LCD
Protection rating	IP21
Parallel connection	Up to 15 units
Output control	ON / OFF switch

03	MECHANICAL
Product dimensions (W x D x H)	460 x 250 x 922 mm
Net weight	132.4 kg
Packing dimensions	980 x 545 x 436 mm
Gross weight	153.4 kg
Installation	Floor-mounted

04	SYSTEM BENEFITS
● Modular architecture for future capacity expansion ● High charge and discharge efficiency ● Designed for solar self-consumption and backup ● Real-time status visibility through the front LCD ● Multi-protocol integration with compatible inverters	

ENGINEERING NOTE

Specifications are reproduced from the supplied M16S314BL-F source sheet. Product configuration, inverter compatibility, certifications and final values must be confirmed in the approved REFU VOLT engineering release before commercial use.